Dylan Burkey

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Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Mechanical Engineering – Manufacturing Concentration Aug 2023 – May 2027
Major GPA: 3.54/4.0 | Dean's List (4 Semesters) | Georgia Tech Europe (Spring 2025)

Experience

Najafi Laboratory – Georgia Institute of Technology Aug. 2026 – Present
Undergraduate Research Assistant *Atlanta, GA*

- Design and fabricate SOLIDWORKS-modeled components, 3D-printed parts, and custom mechanical fixtures for neuroscience experiments; refine geometry and assembly features based on fit, function, and experimental requirements to improve integration and design reliability.
- Analyze behavioral video and imaging datasets using Python-based computer vision and machine learning pipelines (DeepLabCut, Suite2p) to extract pose and neural fluorescence signals, align synchronized time-series data, and prepare datasets for correlation and quantitative analysis.

Amazon May 2026 – Aug. 2026
Area Manager Intern *Miami, FL*

- Optimized automated induct conveyor systems on the CARDINAL robotic workcell; performed sensor alignment, PLC I/O verification, and troubleshooting to increase pallet-stacking throughput 8% and tote first-pass yield 15%.
- Reduced internal freight rollover by 11.3K packages per month by developing a Python/JavaScript package management application; implemented in 5 US fulfilment centers and earned a Top 50 finish at Amazon's AI Intern Summit.
- Coordinated 100+ associates across engineering, maintenance, and operations; documented process improvements and standardized workflows.

Smurfit Westrock May 2025 – Aug. 2025
Manufacturing Engineering Intern *Mooresville, NC*

- Implemented manufacturing improvements across high-speed die-cutter production; analyzed tooling wear, component geometry and process data to create a data-driven maintenance and repair schedule, reducing downtime 25% and extending blade service life 10%.
- Created and validated 2D engineering drawings and 3D CAD models in SOLIDWORKS and ArtiosCAD; applied GD&T per ASME and DFM principles to support product development and manufacturing capability for new packaging products.
- Reduced stockout-related production downtime 35% by developing an inventory management application in Oracle JD Edwards ERP to analyze material and component consumption patterns, supporting maintenance teams and manufacturing operations.

Engineering Projects

Yellow Jacket Aviation - RV-14 Aircraft Build Feb. 2023 – Present

- Manufacturing team lead for RV-14 aircraft kit fabrication, translating build plans into wing and fuselage assembly workflows, resolving tooling and sequencing constraints while accounting for FAA amateur-built requirements.
- Evaluate build facilities, tooling requirements, and component sources to develop BOMs and manufacturing cost estimates; established procurement priorities and startup funding requirements to support a phased aircraft build.

Toothless: Autonomous Robot Aug. 2025 – Dec. 2025

- Designed, fabricated, assembled, and tested an autonomous electromechanical robot integrating machined components, pneumatic actuators, sensors, 3D-printed parts, and laser-cut structures with Arduino-based controls; placed 2nd of 70+ teams.
- Created 3D component models and part specifications in SOLIDWORKS/Onshape; applied GD&T, tolerance stack-up analysis, and DFM/DFA to control interfaces, verify clearances and fits, reduce design risk, and improve manufacturability through iterative prototyping.

Case Western Reserve University - Bearing Vibration Analysis Jan. 2026 – Mar. 2026

- Diagnosed bearing faults using vibration data from a 2-hp motor in MATLAB (RMS, regression, Newton-Raphson); estimated speed within 0.3%.
- Predicted bearing remaining useful life via numerical integration (RK4, Euler); validated within 0.1 hr of true failure time.

Involvement and Leadership

Curious Cardinals Dec. 2023 – Present
Technical Mentor

- Mentored aspiring mechanical engineering students in creating CAD and robotics project portfolios to support professional development.

Kappa Sigma Fraternity Aug. 2024 – Present
Social Chair

- Administered a $40K semester budget, managing vendor negotiations, cash flow tracking, and logistics for 50+ social events for students.

Technical Skills

CAD & Product Definition: SOLIDWORKS, Siemens NX, Onshape, Ansys Granta, GD&T, Engineering Drawings, BOM Development
Manufacturing & Quality: Lean Six Sigma Yellow Belt, DFM/DFA, CAPA, RCA, New Product Introduction (NPI), Process Improvement
Fabrication & Shop: Mill, Lathe, 3D Printing, Laser Cutting, Bandsaw, Drill Press, Power Tools, Precision Measurement, Assembly
Automation & Controls: PLC Fundamentals, PLC I/O Verification, Ladder Logic, Sensors & Actuators, Arduino, Industrial Automation
Programming & Data Analysis: Python (NumPy/Pandas), MATLAB, Minitab, Statistical Analysis, Machine Learning